

Closing the Fidelity Gap: Real-Stack-Grounded Link Abstraction for 5G Digital Twins

Abstract—Network digital twins are essential for developing O-RAN rApps before deployment, yet open platforms trade off protocol-stack realism, site-specific propagation, network scale, and closed-loop control—none combines all four. We address the fidelity gap at its root: the link-to-system (L2S) abstraction. Instead of idealized SINR-to-throughput curves, our twin maps per-resource-element SINR to throughput through a curve *measured on OpenAirInterface’s (OAI) real physical layer* (`nr_dlsim` with LDPC decoding), so network-scale KPIs inherit a real receiver’s implementation losses. We quantify this gap: at 10 dB SINR the OAI curve delivers 2.16 bits/s/Hz—38% below the Shannon bound—and its MCS steps span up to 2.2 dB versus the ~ 1 dB of an idealized staircase. The twin builds multi-cell, multi-user 5G scenes from OpenStreetMap, ray-traces every cell-to-user channel with Sionna RT, and hosts a proportional-fair traffic-steering rApp; three bridges connect it to a live OAI stack, including per-cell channel injection on a two-DU stack. Across three cities and eight random layouts each (24 runs), the rApp raises median per-user throughput by $9.5 \pm 11.3\%$ and served cell-edge throughput by $22.4 \pm 24.8\%$, improves Jain fairness by 9.1%, and cuts peak cell load by 0.75 users, leaving aggregate rate and coverage-limited outage unchanged. The gap is not cosmetic: an idealized curve inflates throughput by 35%, changes the rApp’s steering decisions in 42% of layouts, and mis-estimates its per-deployment gain by a median 7.5 points. Finally, we close the loop on a live OAI 5G Standalone stack and *measure* the gap: at matched bandwidth, link-level rates overpredict delivered goodput by a median of 80%, growing with SINR rather than acting as a constant protocol tax, and decomposing into TDD duty, MAC overhead and the emulator’s own ceiling; through the rApp the twin *under-predicts* per-UE steering gain by a median 12 points. We also report an instructive negative result: on two real cells the rApp’s steering executes as real handovers, but a null control on *identical* cells—where the true effect is zero—shows the natural goodput probe reporting $-21.0 \pm 1.4\%$, retracting a multi-cell gap we previously reported.

Index Terms—Digital twin, 5G NR, OpenAirInterface, link-to-system abstraction, O-RAN, rApp, traffic steering.

I. INTRODUCTION

The digital twin is becoming the environment in which AI/ML-driven RAN control—O-RAN rApps and xApps—is trained and validated before it touches a production network [1]. Its value hinges on data fidelity: a traffic-steering rApp decides which cell serves each user from signal, interference, and load; those quantities depend jointly on 3D propagation (site-specific), on neighbouring cells (multi-cell), and on a link model that turns SINR into a real modulation-and-coding outcome (protocol-stack realism). Evaluating such a controller requires all of these together, and—because random

deployments vary widely—across many layouts rather than a single scenario. A twin reporting one headline number from one scenario can mislead: our own single-seed pilot suggested a uniformly large rApp gain, which a 24-run study tempered into a robust-direction, variable-magnitude result.

Open tooling splits into camps that do not meet. System-level simulators scale to many cells but abstract the physical (PHY) layer with idealized SINR-to-throughput curves. OAI-based emulators execute the real 5G-NR stack but, with realistic propagation, cover a single link. Commercial twins close the rApp loop but are proprietary. No open platform combines a real protocol stack, a site-specific ray-traced channel, a multi-cell network, and a closed control loop (Table I).

A subtler gap concerns the *L2S abstraction* itself. System-level evaluation cannot afford full PHY processing per user, so it relies on a mapping from SINR to block-error-rate (BLER). Open simulators use idealized or 3GPP-calibrated curves; we obtain ours from OAI’s real receiver (`nr_dlsim` with LDPC decoding and rate matching), so network-scale KPIs inherit a real implementation’s losses—particularly at MCS transition boundaries where control decisions are sensitive. At 10 dB SINR, the OAI-measured curve yields 2.16 bits/s/Hz, while the Shannon bound gives 3.46 bits/s/Hz—a 38% gap that an idealized curve would ignore (Fig. 2).

This paper contributes:

- 1) An *OAI-grounded L2S abstraction, and evidence that it matters for control*. The SINR-to-throughput curve is measured from OAI’s real PHY (38% below Shannon at 10 dB; non-uniform MCS steps up to 2.2 dB). Replacing it with an idealized curve changes the rApp’s decisions in 42% of layouts and mis-estimates its per-deployment gain by a median 7.5 points.
- 2) An open, CPU-only, multi-cell/multi-user digital twin: OpenStreetMap geometry $\rightarrow$ Sionna RT ray tracing $\rightarrow$ per-UE/cell throughput, with a swappable network source for real cell-site data.
- 3) Three twin-to-real-stack bridges: (a) exact ray-traced channel injection into a live OAI gNB–UE link, (b) a per-UE interference-as-noise-floor mapping, and (c) *per-cell* ray-traced injection on a two-DU stack, exploiting the UE’s role as rfsimulator server: the k -th DU’s waveform is filtered by `rfsimu_channel_uek` and the DUs’ signals are physically summed, giving combined-waveform inter-cell interference.

- 4) A measured end-to-end fidelity gap, and a null control that retracts a previous one. On one gNB, at matched bandwidth, deliverable goodput sits a median 80% below the twin's link-level rate, growing with SINR (72% $\rightarrow$ 87%); we decompose it into TDD duty, MAC overhead, and the emulator's ceiling. Through the rApp the twin under-predicts per-UE steering gain by a median 12 points. On two real cells, a null control on *identical* cells shows the natural goodput probe reporting $-21.0 \pm 1.4\%$ where truth is 0, retracting our earlier multi-cell gap.
- 5) A multi-scene, multi-seed evaluation (24 runs, 3 cities) of a proportional-fair traffic-steering rApp, reporting variance and outage, a load-sensitivity sweep separating coverage- and interference-limited regimes, and a correlation analysis showing rApp gain is limited by baseline outage rather than raw load imbalance.

II. RELATED WORK

Open RAN digital-twin platforms. Table I positions the closest open efforts on four axes. OWDT [2] pairs OAI with Sionna RT for a faithful single gNB-UE link but is single-cell and monitor-only. Tiny-Twin [3] runs a real OAI stack for multiple UEs with a RIC loop, but single-cell from replayed traces. Orange's RIC-TaaP/ns-O-RAN and OpenTwin [4] close rApp loops across many cells—OpenTwin adds twin-to-real KPM self-correction—but on ns-3's abstracted PHY. Recent OAI+FlexRIC work [5] closes handover loops across real cells over the air, without a ray-traced channel. Commercial twins (e.g. Rimedo Labs' NDT with VIAVI) close the loop but are neither open nor reproducible.

Link-to-system abstraction. System-level simulation [6] relies on L2S mapping: a link-level simulator characterises SINR-to-BLER per MCS, and an effective-SINR mapping (EESM [7], [8]) collapses a frequency-selective SINR vector into an AWGN-equivalent value. Our lookup is measured from OAI's own PHY rather than from idealised curves. Our interference-as-noise-floor bridge is a site-specific generalisation of the 3GPP OFDMA channel noise generator (OCNG) [9]: we derive each UE's interference from the ray-traced neighbour channels.

O-RAN control loops. The O-RAN architecture [10], [11] externalises RAN control: xApps on the near-RT RIC (via E2) and rApps on the non-RT RIC (via O1/A1). Traffic steering via cell-individual offsets (CIO) is a canonical rApp use case [1], which a digital twin lets one validate before deployment.

III. DIGITAL TWIN ARCHITECTURE

The platform is a pipeline with two consumers of one physics substrate (Fig. 1): a fast analytical layer covering every cell and UE, and a real-stack layer grounding sampled links in OAI. Both are driven by the same ray-traced channel, and the network data source is swappable so synthetic and real deployments share one path.

Geometry. A WGS84 bounding box drives an OpenStreetMap (Overpass) query; footprints are extruded into a 3D Mitsuba scene with ITU materials.

TABLE I
OPEN RAN DIGITAL-TWIN PLATFORMS ON FOUR AXES.

Platform	Real stack	Site RT	Multi cell	Closed loop
OWDT [2]	✓	✓	×	×
Tiny-Twin [3]	✓	×	×	✓
Orange ns-O-RAN / RIC-TaaP	×	✓	✓	✓
OpenTwin [4]	×	×	✓	✓
OAI+FlexRIC [5]	✓	×	✓	✓
OAI VRTSim / RT emulator	✓	✓	×	×
This work	✓	✓	✓	✓*

^areplayed traces; ^bns-3 abstract PHY; ^cover-the-air; ^dmulti-node on roadmap; *SA loop closed & measured on live stack (scripted F1 HO for multi-cell).

Network. Cells and UEs come from a swappable source: a seeded random generator (N sites $\times$ 3 sectors, uniform UE drops) for controlled experiments, or a CSV reader for real cell-site coordinates (e.g., OpenCellID). Both feed the same pipeline, so the twin can be calibrated on synthetic data and run on real site data without code changes.

Channel. Sionna RT [12] places one transmitter per cell (3GPP sector pattern, planar array, downtilt) and one receiver per UE and computes the channel frequency response for every cell-to-UE link over an OFDM resource grid, capturing reflections, diffractions, and scattering from the building geometry. Ray-tracing depth is 3, balancing fidelity against CPU runtime.

IV. OAI-GROUNDED KPI ENGINE

The KPI engine is where the twin's fidelity is grounded. Let $\mathbf{H}_{u,c}(k) \in \mathbb{C}^{N_r \times N_t}$ be the ray-traced channel of link $c \rightarrow u$ on resource element (RE) k , and P_c^{tx} the cell transmit power. The wideband received power is $P_{u,c}^{\text{rx}} = P_c^{\text{tx}} \|\overline{\mathbf{H}_{u,c}}\|_F^2$, where the bar denotes averaging over antennas and REs. Each UE associates to the cell maximising received power biased by a cell-individual offset (CIO), the O-RAN control knob:

$$s(u) = \arg \max_c P_{u,c}^{\text{rx}} \cdot 10^{\text{CIO}_c/10}. \quad (1)$$

Because each cell serves one UE at a time under airtime sharing, the optimal single-stream precoder is the matched filter, so the serving per-RE gain is the largest singular value $\sigma_{\max}$ of the serving channel. With flat per-RE transmit power p_c , a neighbour load factor $\rho \in [0, 1]$, and thermal noise $N_0 = kTB \cdot \text{NF}$, the per-RE SINR is

$$\gamma_u(k) = \frac{p_s \sigma_{\max}(\mathbf{H}_{u,s}(k))^2}{\rho \sum_{c \neq s} \frac{p_c}{N_t} \|\mathbf{H}_{u,c}(k)\|_F^2 + N_0}. \quad (2)$$

The per-RE SINRs are compressed to one effective SINR by exponential effective SINR mapping (EESM), which preserves the frequency selectivity the ray tracer produces:

$$\gamma_u^{\text{eff}} = -\beta \ln \left(\frac{1}{K} \sum_k e^{-\gamma_u(k)/\beta} \right), \quad (3)$$

with β set per modulation order (QPSK, 16QAM, 64QAM) following established link-abstraction literature [8], and a single `eesm_beta_scale` parameter for calibration against OAI ground truth.

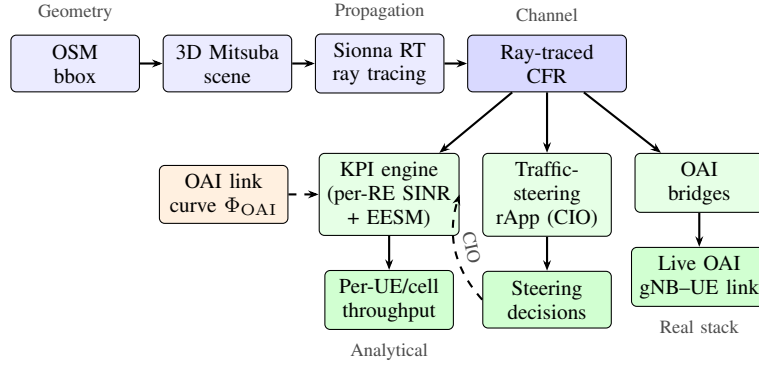


Fig. 1. Architecture. OpenStreetMap geometry and a swappable network source feed Sionna RT; the ray-traced channel drives (left) an analytical multi-cell KPI engine and the traffic-steering rApp, and (right) a live OAI link over the exact channel with a per-UE interference noise floor. The OAI-measured link curve Φ_{OAI} grounds the analytical throughput in a real receiver.

OAI-measured link curve. The effective SINR is mapped to a modulation-and-coding scheme (MCS) and spectral efficiency η_u through a curve Φ_{OAI} *measured offline from OAI's nr_dlsim* (real LDPC decoding and rate matching), i.e. $(m_u, \eta_u) = \Phi_{\text{OAI}}(\gamma_u^{\text{eff}})$. This is the key distinction from prior system-level twins: the mapping carries a real receiver's implementation losses—LDPC decoding margins, rate-matching overhead, and the non-uniform SINR spacing between MCS transitions. It is measured once by sweeping $\text{SNR} \times \text{MCS}$ through *nr_dlsim* and recording the lowest SNR at which first-transmission BLER stays within the 10% target; the resulting staircase (Fig. 2) is a monotone lookup table indexed by effective SINR.

Quantifying the implementation loss. Fig. 2 compares the OAI-measured curve against the Shannon bound and an idealized uniform staircase. At 10 dB SINR the OAI curve delivers 2.16 bits/s/Hz—38% below Shannon—and in absolute terms the loss grows to ~ 1.6 bits/s/Hz at 64QAM as LDPC and rate-matching overhead accumulate. Critically, the MCS transition spacing is non-uniform: the QPSK $\rightarrow$ 16QAM boundary spans 2.2 dB against ~ 1 dB for adjacent QPSK steps. A uniform 1 dB staircase misplaces this transition by over 1 dB, assigning the wrong MCS to boundary users—exactly the users a traffic-steering rApp aims to help.

On a static full-buffer snapshot, proportional-fair scheduling reduces to equal airtime, so the per-UE throughput is

$$R_u = \frac{B_s}{K_s} \eta_u (1 - \text{BLER}_{\text{tgt}}), \quad (4)$$

where B_s is the serving cell bandwidth and K_s its number of attached UEs.

V. TRAFFIC-STEERING RAPP

The rApp adjusts the CIO vector to maximise the proportional-fair utility $U(\text{CIO}) = \sum_u \log R_u$ (Alg. 1). Because U is piecewise-constant in the CIOs (it changes only when a UE's serving cell flips), a fixed small step stalls on plateaus; we therefore line-search each cell's CIO over a grid (coordinate ascent), which is monotonically non-decreasing in

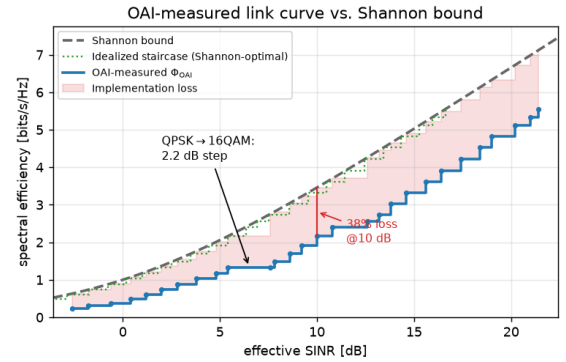


Fig. 2. OAI-measured link curve Φ_{OAI} vs. Shannon bound and an idealized (Shannon-optimal) staircase. The shaded region is the implementation loss (38% at 10 dB). Non-uniform MCS step spacing reflects a real LDPC decoder.

Algorithm 1 Proportional-fair traffic steering

- 1) $\text{CIO} \leftarrow \mathbf{0}$; evaluate R_u and U via Eqs. (1)–(4).
- 2) **repeat** (sweep over cells):
 - a) for each cell c : line-search CIO_c over a grid (others fixed) and keep the value maximising U .
- 3) **until** no cell's line-search improves U (local optimum).

U by construction. Raising a cell's CIO pulls boundary UEs in (absorbing load); lowering it sheds them to neighbours.

Every KPI evaluation reuses the same engine with a `cio_db` vector, so the control loop rides on the exact same twin used for the baseline. The rApp's output is a set of per-UE handover decisions that a real stack would execute via O1/telnet. The coordinate-ascent structure guarantees monotonic improvement in U and converges in 2–3 sweeps, suitable for the non-RT RIC timescale.

VI. TWIN-TO-REAL-STACK BRIDGES

Three bridges connect the analytical twin to a live OAI stack, grounding per-link and per-cell realism.

Channel injection. A bridge converts a link's ray-traced channel frequency response to a sample-spaced impulse response (IFFT), retains the strongest causal

TABLE II
RADIO AND TOPOLOGY PARAMETERS.

Parameter	Value
Carrier / bandwidth	3.5 GHz / 20 MHz
Subcarrier spacing	30 kHz (numerology 1)
Cell transmit power	46 dBm
BS array / UE antennas	$4 \times 1 / 1$
Receiver noise figure	7 dB
Ray-tracing interaction depth	3
Cells / UEs per layout	9 / 30
Default inter-cell load ρ	1.0

taps, normalises them to unit energy, and maps each tap delay to a sample index. These taps override the rfsimulator’s channel through a patch to OAI’s channel module (`oai_rt_inject_channel` in `random_channel.c`). When `OAI_RT_TAPS` is set, the gNB–UE link runs over the site-specific channel. Injection is confirmed in the gNB log (`[RT] injected 4 taps ... channel_length=49`); the link then runs its PHY/MAC over the ray-traced channel.

Interference-as-noise-floor. A live single-gNB rfsimulator link has no notion of the other cells a UE hears. A second bridge computes, per UE, the inter-cell interference and expresses it as a noise-floor rise $\Delta N_u = 10 \log_{10}((I_u + N_0)/N_0)$, where I_u is the load-scaled sum of non-serving cells’ received power. This is written as a per-client OAI noise power, so a single-link run reflects its multi-cell context—a site-specific generalisation of OCNG [9].

Per-cell channel injection. With two DUs the noise-floor proxy is unnecessary. In the handover topology the UE is the rfsimulator *server*, so the k -th DU to connect is filtered by channel model `rfsimu_channel_uek` and `rxAddInput` accumulates every connection into the UE’s receive buffer. One taps block per DU therefore presents each cell’s own ray-traced link gain, and the non-serving cell’s waveform is summed in as genuine co-channel interference. We also make DU1 co-channel with DU0 (identical but for PCI/cell identity); the stock configuration places them on different carriers.

VII. EXPERIMENTAL SETUP

Scenes. Berlin-Mitte (2,623 buildings), Paris-Opéra (121), Manhattan-Midtown (119); geometry fetched/built once per scene and reused across seeds.

Random layouts. Eight seeded layouts per scene—24 runs (323 s on one CPU server). Radio and topology parameters are summarised in Table II.

Configuration sweep. Neighbour load factor in $\{0, 0.25, 0.5, 0.75, 1.0\}$ on a cached channel per scene.

Metrics. Per-UE downlink throughput (median, mean); *served cell-edge* rate, the 5th percentile among UEs with non-zero throughput (a raw 5th percentile collapses to zero once outage exceeds 5%); *outage*, the fraction of UEs with zero throughput; Jain’s fairness index; peak cell load. rApp gains are mean $\pm$ std over the eight seeds per scene, pooled across all 24 runs; undefined when a baseline metric is zero.

TABLE III
RAPP GAINS VS. STRONGEST-CELL (MEAN $\pm$ STD OVER SEEDS).

Metric	Overall	Berlin	Paris	Manh.
Median tput gain	$9.5 \pm 11.3\%$	10.5%	4.5%	13.3%
Served edge gain	$22.4 \pm 24.8\%$	41.0%	6.0%	20.2%
Jain fairness gain	$9.1 \pm 11.0\%$	11.2%	4.4%	11.5%
Peak load (UEs)	−0.75	−0.88	−0.50	−0.88
Mean tput gain	$0.5 \pm 5.8\%$	−0.7%	−0.3%	2.5%
Outage change	0.0 pts	0.0	0.0	0.0

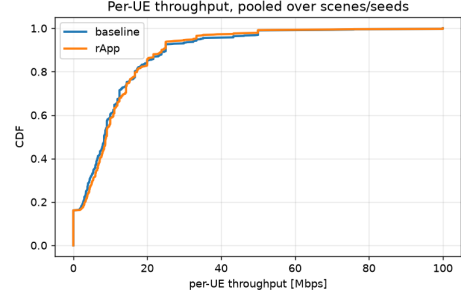


Fig. 3. Per-UE throughput CDF, baseline vs. rApp (pooled over all scenes and seeds). The rApp lifts the low-to-median part of the distribution.

VIII. RESULTS

A. Traffic-steering rApp

Across 24 runs, relative to strongest-cell association (Table III, Fig. 3): the rApp helps the users a strongest-cell policy under-serves (median, cell-edge, fairness) and sheds peak load. In absolute terms the overall median per-UE rate rises from 8.5 to 9.2 Mbps and the served cell-edge rate from 3.5 to 4.2 Mbps, while the mean is essentially unchanged (11.4 Mbps). Three honest observations: aggregate rate barely moves (+0.5%); the proportional-fair objective trades sum for fairness), outage is unchanged (steering relocates covered users, it does not create coverage), and variance is high (layout-dependent)—motivating multi-seed evaluation.

The per-scene breakdown is instructive (Table III). Berlin, the densest scene, shows the largest served cell-edge gain (+41%): dense blockage produces stronger load imbalance for the rApp to exploit. Paris shows the smallest gain (+6%) and the highest outage (23.8%): when many users lack coverage entirely, steering has little covered load to rebalance. Manhattan sits in between. Across all three cities the direction is robust—median, cell-edge, and fairness improve while peak load falls—while the magnitude is strongly deployment-dependent, exactly what a single-scenario study would misrepresent.

B. The fidelity gap changes rApp evaluation

The 38% link-curve gap of Sec. IV is not cosmetic—it changes what the twin tells an rApp developer. We re-run the *identical* rApp on a twin whose *only* change is the link curve: Φ_{OAI} is replaced by an idealized curve assigning the same MCS set at the Shannon-optimal SINR, with channel, association and interference held fixed. The idealized twin inflates

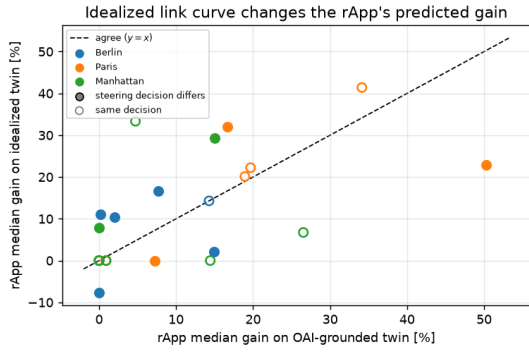


Fig. 4. Predicted rApp median gain on an idealized-curve twin vs. the OAI-grounded twin (24 layouts). Points off $y=x$ show the idealized twin mis-estimating the gain; filled markers are the 10/24 layouts where it also selects different steering actions.

baseline median throughput by 35%. More consequentially, it *disagrees on the control decision*: across the 24 layouts its predicted rApp median gain deviates from the OAI-grounded value by a median 7.5 (mean 8.2, max 29) points, the two agree only moderately ($r=0.59$; Fig. 4), and in 10/24 layouts (42%) the rApp selects a *different* set of steering actions (mean Jaccard 0.75). Yet the *ensemble-average* gain is nearly identical (10.9% vs 10.4%); an idealized twin looks calibrated in aggregate while being unreliable per deployment—exactly the regime in which an operator ranks a policy for a specific site.

C. The real fidelity gap

The analysis above compares two *models*; we now close the loop on the real stack and measure the gap directly. We stand up a full OAI 5G Standalone system on the same CPU host—core (AMF/SMF/UPF), gNB, and a UE that registers, establishes a PDU session and carries user-plane traffic. On a single gNB we ground the *link* dimension: we pin the downlink operating point by capping the scheduler MCS over a clean channel, sweep the cap across MCS 0–28, and measure end-to-end goodput at each fixed MCS, which indexes the same OAI curve the twin uses (a shared SINR axis). Two details decide the answer. The live cell is 106 PRB at 30 kHz SCS, i.e. $B = 38.16$ MHz occupied, not the twin scenario’s 20 MHz, so the twin’s rate must be evaluated at the cell’s own bandwidth. And a saturating UDP probe is unusable: in reverse mode `iperf3` reports the *sending* rate, so with the scheduler pinned to MCS 4 (a 9.8Mbps MAC) an 8Mbps offer reads exactly 8.00Mbps at $\sim 0\%$ loss. TCP cannot exceed the channel but needs tens of seconds to leave slow start, so we settle and measure its plateau.

The twin’s link-level rate $B\eta_{\text{OAI}}(1 - \text{BLER})$ overpredicts delivered goodput by a median of 80% (Fig. 5, left). The gap is *not* a constant protocol overhead: it grows from 72% at MCS 0 to 87% at MCS 28, and the bandwidth implied by $\text{goodput}/\eta_{\text{OAI}}$ falls from 10.7 to 5.0 MHz—the signature of a throughput ceiling. The MAC’s scheduled throughput separates the causes: at MCS 28 the twin predicts 190.8Mbps,

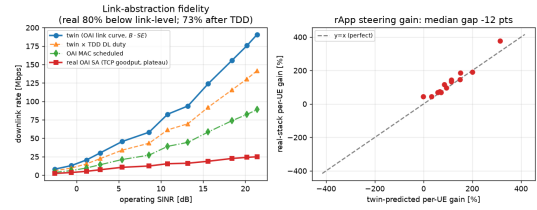


Fig. 5. Measured real fidelity gap (single gNB). Left: at matched operating SINR—the twin’s link-level rate, that rate scaled by the TDD DL duty, OAI’s scheduled MAC throughput, and delivered goodput; the spread between the last two is the emulator’s ceiling. Right: per-UE rApp steering gain, twin vs. measured curve (8 layouts, 13 steered UEs).

the TDD DL duty (104/140 symbols) allows 141.7, OAI’s MAC schedules 88.9 once DMRS, PDCCH and HARQ are paid, and the application receives 24.9. Only the first step is a modelling choice—the last is the rfsimulator’s real-time ceiling, an emulator artefact rather than a property of 5G, which we report as such rather than fold into “the fidelity gap”.

Propagating both curves through the rApp over eight layouts (240 per-UE evaluations, 13 steered UEs) gives Fig. 5 (right). We keep the twin’s airtime share B_c/K_c in *both* rates—that term is most of the rApp’s benefit—so the residual isolates the link abstraction. The twin predicts a mean per-UE gain of 111% against a measured-curve gain of 130%: it *under*-predicts by a median 12 points (mean -18.7 , std 21.3; we do not headline a mean whose spread exceeds it). Because the measured curve is flatter, a UE pushed onto a weaker cell loses less rate than the twin believes, so load balancing pays more, not less.

D. Multi-cell closed loop, and what it does not yet measure

Two real cells (CU plus two F1 DUs, PCI 0/1) match a two-sector twin (20 UEs). The rApp’s association decision executes as a real inter-cell handover via scripted CU telnet (`ci_trigger_fl_ho`), with the PDU session preserved and the flow continuing on the target cell: the loop is closed. *Measuring* the delivered gain is another matter. Our first attempt reported the twin’s per-UE gain (66%) overshooting the delivered gain (9.0%) by 57 ± 165 points, and blamed the DUs’ clean channel. That does not survive a null control: give the cells the *same* channel and the true effect is exactly zero, so whatever the harness reports is instrument bias. It reports $-21.0 \pm 1.4\%$ (Fig. 6). The cause is timing—a 6 s TCP flow started 3 s after each handover, inside both slow start and the link-adaptation ramp—and because `trigger_fl_ho` always moves DU0→DU1, measurement order and cell identity were perfectly confounded, so the reported sign followed the direction the rApp happened to choose (on identical cells the probe yields a mean DL MCS of 16.5 on PCI 0 against 10.7 on PCI 1). Probing at TCP’s plateau gives $-2.4 \pm 2.9\%$, consistent with zero, with MCS 28 on both. We withdraw the 57 ± 165 number.

With the instrument fixed we ground the channel (Sec. VI). Injection works: for every layout the UE log confirms both

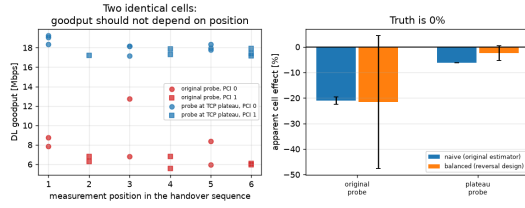


Fig. 6. Null control: both cells carry the *same* clean channel, so the true cell effect is 0%. Left: with the original probe, goodput depends on position in the handover sequence, and PCI 0—measured at odd positions—separates from PCI 1. Right: the original estimator reports $-21.0 \pm 1.4\%$; the plateau probe gives $-2.4 \pm 2.9\%$.

per-cell models carrying exactly the twin’s ray-traced gains (-0.00 and -5.87 dB for one layout’s 5.87 dB split), and the UE reaches MCS 27 on the twin’s stronger cell. Two obstacles remain, neither of them the twin’s. First, with channel models on both rfsim connections the loop destabilises: over three layouts, two F1 handovers never complete and the third only through 70 UE re-synchronisations, so its goodput measures the emulator’s recovery, not the channel. Second, rfsim’s noise floor is a config knob, not $kTB \cdot NF$: injection sets the *relative* per-cell gain but not the absolute SINR. Sweeping the floor from -22 to -2 dB with one cell muted, the operating MCS goes 8, 19, 14, 18, 14, 0—non-monotone, so the map cannot be inverted to place a cell at a target operating point. Meanwhile, at the idle-neighbour condition a one-UE stack presents, the twin puts both cells at MCS 28 and predicts *no* link-level difference in two of three layouts. A controlled multi-cell gap needs absolute operating-point matching, which this emulator cannot deliver; we report the negative result rather than a number.

E. Coverage, outage, and load sensitivity

Mean UE outage is 16.2% at full load (Berlin 8.8%, Manhattan 16.3%, Paris 23.8%); rApp median gain declines with baseline outage ($r = -0.42$) more than with load imbalance ($R^2 = 0.04$). Lowering neighbour load from 1.0 to 0 raises mean throughput 1.7 – $1.8\times$, separating a coverage-limited tail steering cannot fix from an interference-limited body where load-balancing acts.

IX. DISCUSSION AND LIMITATIONS

We state these plainly. (1) The OAI Standalone loop is closed on one gNB (link calibration, Sec. VIII-C) and on two cells with scripted F1 handovers (Sec. VIII-D); autonomous E2-RC steering remains future work. (2) The single-gNB gap is measured against *this* emulator, which delivers only 24.9 of the 88.9 Mbps its own MAC schedules at MCS 28; the error attributable to the twin is bounded by the MAC line, not the goodput line—why Fig. 5 plots both. (3) Per-cell injection is verified and rfsim sums the DUs’ waveforms—so inter-cell interference is physically combined, superseding the noise-floor rise ΔN_u the analytical engine still uses—but the multi-cell measurement does not close (Sec. VIII-D), and the injected taps carry each link’s gain, not its delay profile

(3.84 MHz of exported CFR resolves delays only to 260 ns; the same sampling limits the EESM). (4) The evaluation is a static full-buffer snapshot, single carrier, downlink; mobility, FFR and uplink are natural extensions, and rApp benefits are high-variance and layout-dependent (we report full distributions). (5) Validation is stack-versus-model; field calibration and real cell-site databases are the next steps.

X. CONCLUSION

We presented an OAI-grounded, site-specific multi-cell twin with a traffic-steering rApp, three real-stack bridges, and a multi-scene evaluation. Replacing the OAI curve with an idealized one changes steering in 42% of layouts (median gain error 7.5 points). On the real OAI stack, deliverable goodput is a median 80% below the twin’s link-level prediction, growing with SINR and decomposing into TDD duty, MAC overhead and the emulator’s own ceiling; through the rApp the twin under-predicts per-UE steering gain by a median 12 points. Extending the loop to two real cells, a null control on *identical* cells shows the natural goodput probe reporting a $-21.0 \pm 1.4\%$ cell effect where truth is zero, retracting a multi-cell gap we previously reported. An OAI-grounded twin must be closed against the real stack before its control conclusions can be trusted—and the loop itself must be closed against a null control before *its* conclusions can be. Code, data and OAI logs reproduce every number (Apache-2.0).

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